

Spring 20 Techniques Exam practice

The problems collected here are given as practice for the exam. These are *not* meant to be comprehensive and in order to do well you should study as many problems as possible. In particular, the problems that we have already had on recent exams and practice exams. Also, don't forget quiz problems!

1. Find $\int \tan^3(x) \sec^4(x) dx$.

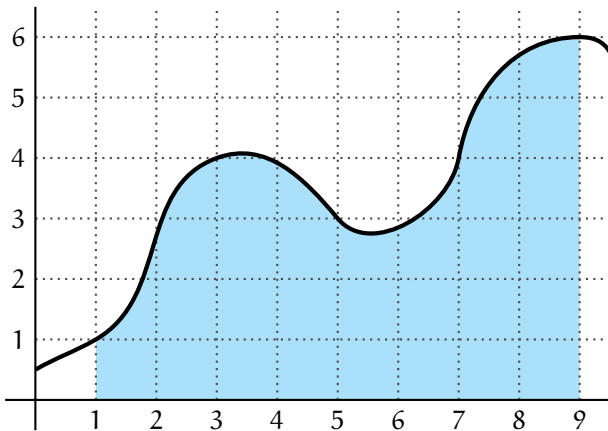
2. Find $\int_1^{\infty} \frac{1}{t \ln(e^2 t) \ln(e^4 t)} dt$.

3. Find $\int \cos^3(x) \sin(\sin(x)) dx$.

4. Find $\int_{-\infty}^0 \frac{e^{3t}}{(4 - e^{2t})^{3/2}} dt$

5. Use the picture of the curve $y = f(x)$ below to do the following:

- (a) Estimate the area under $f(x)$ for $1 \leq x \leq 9$ using the trapezoid rule with 4 equal intervals.
- (b) Estimate the area under $f(x)$ for $1 \leq x \leq 9$ using Simpson's rule with 4 equal intervals.



6. Find the following improper integral $\int_0^1 \frac{e^{\sqrt[3]{x}}}{\sqrt[3]{x}} dx$.
Indicate any limits needed.

7. Find all $b > 0$ such that $\int_1^{\infty} \frac{b}{x^2 + b^2} dx = 1$.

8. Find $\int_0^1 9\sqrt{z} \ln(1+z) dz$.

9. Find $\int \frac{1}{(y^2 + 5y + 6)^2} dy$.

10. Recall that the error in Simpson's rule for $f(x)$ over the interval $a \leq x \leq b$ is bounded above by

$$\frac{M(b-a)^5}{180n^4},$$

where M is max of $f^{(4)}(x)$ for $a \leq x \leq b$. Use this error estimate to bound the maximum error for the integral of $f(x) = \sin(2x)$ using Simpson's rule for the interval $0 \leq x \leq \frac{1}{2}\pi$ with $n = 20$ intervals. (You do NOT need to simplify your answer.)

11. Find $\int_0^{2\pi} 64 \sin^4(x) \cos^4(x) dx$.

12. Find $\int \frac{du}{\sqrt{u}\sqrt{u-6}(u-3)^3}$.

13. Find $\int e^t \cos(3t) dt$.

14. Find $\int \tan(\theta) \sin^3(\theta) d\theta$.